Department of Computer Science

COMP233 – Discrete Mathematics

First Semester 2026/2027 (1261)

Quiz 1 | Chapters 2–3 | Form D

Time: 30 minutes | **Total:** 100 marks | 3 questions

Name: _____ **ID:** _____ **Section:** ____

Instructions: Answer all three questions on the answer sheets only. Show all requested steps and justifications. In this quiz, $\sim$ means negation.

Question 1 [20 marks]

(a) Construct a **complete truth table** for

$$H = (p \to q) \to (\sim p \to \sim q).$$

Classify H as a tautology, a contradiction, or neither. Justify from your table. *[15 marks]*

(b) For this part, let $p =$ "The backup is complete" and $q =$ "The files can be restored." Formalize each statement separately using p and q. *[5 marks; 2.5 each]*

(i) A complete backup is sufficient for the files to be restored.

(ii) A complete backup is necessary for the files to be restored.

Question 2 [35 marks]

(a) Simplify $(\sim p \to q) \wedge (\sim p \to \sim q)$ as far as possible using logical equivalences. **Name the law used at each step.** Do not use a truth table. *[15 marks]*

(b) From the premises $p \vee q$, $q \to r$, and $\sim r \wedge s$, derive p using **rules of inference**. Number your lines and state each rule and its supporting lines. *[20 marks]*

Question 3 [45 marks]

For parts **(a)**–**(b)**, the domain L is all items in a library. Let $R(x) =$ "x is reserved" and $D(x) =$ "x is digital." Consider:

"No reserved item is digital."

(a) Formalize the statement in FOL. *[10 marks]*

(b) Write the negation of the statement in symbolic form. Apply negations only to individual predicates. *[10 marks]*

For parts **(c)**–**(d)**, let $U = \{\text{Ala}, \text{Basma}\}$ and $F = \{f_1, f_2, f_3\}$. The table gives all truth values of $A(u, f)$, meaning "user u can access file f." Consider the statement S below.

$A(u,f)$	f_1	f_2	f_3
Ala	T	F	T
Basma	F	T	T

$$S: \quad \forall u \in U \, \exists f \in F \sim A(u, f).$$

(c) Translate S into English. Determine its truth value in the table and justify. *[15 marks: translation 5; truth value and justification 10]*

(d) Write the negation of S in symbols. Move negations inside the quantifiers and simplify all double negations. *[10 marks]*

End of question sheet. Use the answer sheets only.

Answer Sheet – Question 1

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Name: _____ **ID:** _____ **Section:** ____

Question 1 [20 marks]

(a) Complete truth table, classification, and justification *[15 marks]*

(b) Separate formalizations for (i) and (ii) *[5 marks]*

Answer Sheet – Question 2

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Name: _____ **ID:** _____ **Section:** ____

Question 2 [35 marks]

(a) Simplification with a named law at each step *[15 marks]*

(b) Numbered inference proof with rules and references *[20 marks]*

Answer Sheet – Question 3

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Name: _____ **ID:** _____ **Section:** ____

Question 3 [45 marks]

(a) Formalization *[10 marks]*

(b) Symbolic negation *[10 marks]*

(c) English translation, truth value, and justification *[15 marks]*

(d) Simplified symbolic negation *[10 marks]*